Minimum wages and employment

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1 Research question

This project replicates Tables 2 to 4 and Figure 1 of Card and Krueger (1994). It asks whether the increase in New Jersey's minimum wage from $4.25 to $5.05 in April 1992 reduced employment at fast-food restaurants. Pennsylvania, where the minimum wage stayed at $4.25, is the comparison group.

A higher minimum wage raises labour costs, so firms may hire fewer workers. However, the effect may differ when employers have wage-setting power or adjust prices and staffing in other ways. The data are used to examine the employment response rather than assume its direction.

2 Data and variables

The data are public.csv from course Package 01. They cover 410 restaurants in four chains, surveyed before and after the policy. The main sample counts are shown below.

Sample	Restaurants
All restaurants	410
New Jersey	331
Pennsylvania	79
Employment observed in both waves	384
Table 4 and main extension sample	357

Full-time-equivalent employment (FTE) equals full-time employees plus managers plus half the number of part-time employees. Missing components are kept missing. Follow-up FTE is zero for six permanent closures. Four temporary closures are missing in the baseline and zero in a sensitivity check.

Table 4 requires employment and wage data in both waves, with an exception for permanent closures, whose follow-up employment is zero. All models use the same 357 stores. Different sample rules explain why its estimate differs from the 384-store comparison.

Meal price is the sum of soda, fries and entree prices. The full-time percentage in Table 2 follows the numerical convention 100 x EMPFT / FTE, rather than an ordinary headcount share. Initial wage gap is max[(5.05 - initial wage) / initial wage, 0] in New Jersey and zero in Pennsylvania.

3 Method

The difference-in-differences (DID) estimate is the employment change in New Jersey minus the change in Pennsylvania. Its causal interpretation assumes the two groups would have had similar employment trends without the policy. There is only one pre-policy survey, so this assumption cannot be checked with pre-trends.

Table 4 regresses the change in FTE on the New Jersey indicator or the initial wage gap, adding chain, ownership and region controls as specified in the paper. Standard errors in the replication table use the conventional OLS formula. HC1 robust standard errors are also checked.

4 Replication results

Table 2 Means of key variables

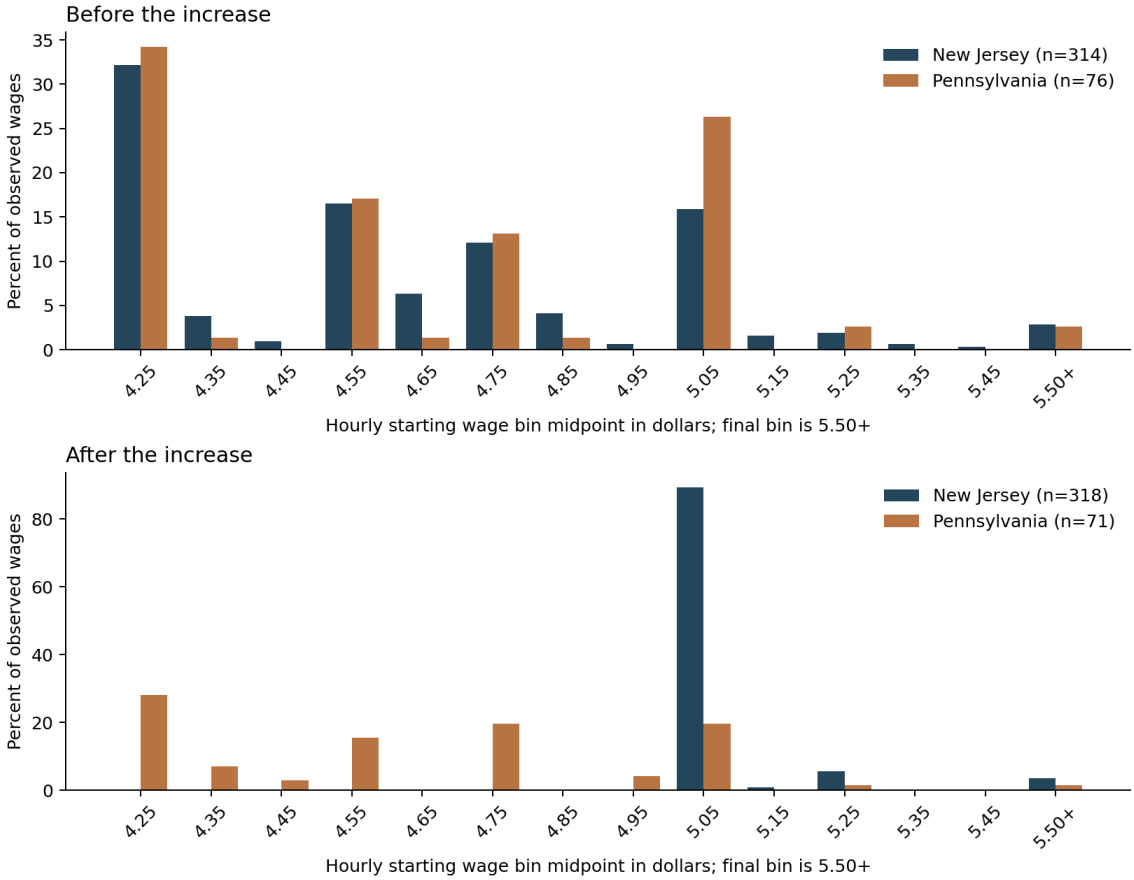
Variable	New Jersey	Pennsylvania	t statistic
1a Burger King	41.09 (2.71)	44.30 (5.62)	-0.52
1b KFC	20.54 (2.22)	15.19 (4.06)	1.16
1c Roy Rogers	24.77 (2.38)	21.52 (4.65)	0.62
1d Wendys	13.60 (1.89)	18.99 (4.44)	-1.12
1e Company owned	34.14 (2.61)	35.44 (5.42)	-0.22
2a FTE employment	20.44 (0.51)	23.33 (1.35)	-2.00
2b Full time percentage	32.85 (1.33)	35.04 (2.73)	-0.72
2c Starting wage	4.61 (0.02)	4.63 (0.04)	-0.40
2d Wage equals 4.25	30.51 (2.53)	32.91 (5.32)	-0.41
2e Full meal price	3.35 (0.04)	3.04 (0.07)	3.96
2f Hours open	14.42 (0.15)	14.53 (0.33)	-0.29
2g Recruiting bonus	23.56 (2.34)	29.11 (5.14)	-0.98
3a FTE employment	21.03 (0.52)	21.17 (0.94)	-0.13
3b Full time percentage	35.87 (1.38)	30.38 (2.81)	1.75
3c Starting wage	5.08 (0.01)	4.62 (0.04)	10.82
3d Wage equals 4.25	0.00 (0.00)	25.32 (4.92)	-5.14
3e Wage equals 5.05	85.50 (1.94)	1.27 (1.27)	36.38
3f Full meal price	3.41 (0.04)	3.03 (0.07)	5.08
3g Hours open	14.42 (0.15)	14.65 (0.33)	-0.65
3h Recruiting bonus	20.32 (2.27)	23.38 (4.85)	-0.57

Standard errors are in parentheses. Prefix 1 refers to store types, 2 to the first survey and 3 to the second survey. Store types, wage thresholds and recruiting bonuses are percentages. Wages and meal prices are dollars. Sample sizes vary by variable and are listed in outputs/tables/table2.csv. The t statistics use unequal-variance standard errors.

Average starting wages rose from $4.61 to $5.08 in New Jersey, while Pennsylvania wages changed little. Average FTE rose slightly in New Jersey and fell in Pennsylvania.

Most means are close to the original table. One difference is the follow-up New Jersey share at exactly $5.05: the supplied file gives 85.50 percent, compared with 85.2 in the paper. Wage-threshold percentages use all state records as the denominator; missing wages are not counted at the threshold. The data are not changed to force a match.

Figure 1 Starting wage distributions



After the policy, New Jersey wages are concentrated near $5.05. The figure excludes missing wages, uses 10-cent intervals from $4.20, and groups wages of $5.50 or more together. Small differences from the original graph can follow from these bin choices.

Table 3 Employment before and after the policy

Statistic	PA	NJ	NJ - PA	NJ low	NJ mid	NJ high	Low - high	Mid - high
Before all available	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	-2.69 (1.37)	-2.17 (1.41)
After all available	21.17 (0.94)	21.03 (0.52)	-0.14 (1.08)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.66 (1.44)	0.74 (1.27)
Change in means	-2.17 (1.22)	0.59 (0.47)	2.75 (1.31)	1.32 (0.81)	0.87 (0.67)	-2.04 (0.98)	3.36 (1.27)	2.91 (1.19)
Change balanced	-2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.20 (0.82)	0.71 (0.69)	-2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
Change temporary zero	-2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	-2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

FTE units; standard errors in parentheses. Low, middle and high refer to initial NJ wages of $4.25, $4.26 to $4.99, and at least $5.00. The available-case change-in-means SE retains within-store covariance; its NJ - PA value is 1.31 versus 1.36 in the original. The balanced-sample SE is 1.34.

In the balanced sample, New Jersey employment increases by 0.47 FTE and Pennsylvania employment falls by 2.28 FTE. The DID is therefore 2.75 FTE. Treating temporary closures as zero lowers it to 2.51. Table 3 point estimates differ from the printed values by no more than 0.01 at the displayed precision.

Table 4 Models for the change in employment

Variable	Model 1	Model 2	Model 3	Model 4	Model 5
New Jersey indicator	2.33 (1.19)	2.30 (1.20)	-	-	-
Initial wage gap	-	-	15.65 (6.08)	14.92 (6.21)	11.98 (7.42)
Chain and ownership	No	Yes	No	Yes	Yes
Region controls	No	No	No	No	Yes
Regression SE	8.79	8.78	8.76	8.76	8.75
Controls joint F test p	-	0.34	-	0.44	0.40
Observations	357	357	357	357	357

The dependent variable is the change in FTE. All models include an intercept. Chain controls omit Burger King. Region controls are SOUTHJ, CENTRALJ, PA1 and PA2, with northern New Jersey omitted. Conventional OLS standard errors are in parentheses.

The estimated NJ effect is 2.33 FTE without controls and 2.30 with chain and ownership controls. The wage-gap coefficient in model 3 is 15.65. For a gap of 0.10, this corresponds to about 1.57 FTE, not 15.65 FTE.

Main coefficient	Original paper	This project
Model 1	2.33	2.33
Model 2	2.30	2.30
Model 3	15.65	15.65
Model 4	14.92	14.92
Model 5	11.91	11.98

The first four coefficients match the original table at two decimal places. Model 5 gives 11.98 (7.42), compared with 11.91 (7.39). The remaining difference is not fully explained, and no observations are changed to remove it.

The model 1 p value is 0.052 with conventional standard errors and 0.110 with HC1 errors. The estimate is positive, but it is not significant at the 5 percent level under either method. The sign should therefore be distinguished from the strength of the evidence.

Checks

The code checks the original file checksum, sample counts and closure rules. It keeps both restaurants with SHEET number 407 because they are in different states. It also checks that the DID equals the corresponding OLS slope and verifies the baseline slope and standard error using SciPy. A run from a separate clean-input folder reproduces the same output files.

5 Extension on company ownership

The extension asks whether company-owned and franchised restaurants respond differently. Company-owned stores may have different hiring decisions or access to funding. I test this by adding NJ x Company to a model with the NJ indicator, company ownership and chain controls. The interaction is the difference between the two ownership groups' DID estimates.

Comparison	Estimate	HC1 SE	95 percent interval	p
Franchise NJ minus PA	2.64	2.08	[-1.46, 6.74]	0.206
Company NJ minus PA	1.70	1.54	[-1.33, 4.73]	0.270
Company minus franchise difference	-0.94	2.58	[-6.01, 4.13]	0.715

The franchise DID is 2.64 and the company-owned DID is 1.70. Their difference is -0.94 FTE, with p = 0.715. There is no clear evidence of a difference. The interval is wide, so this does not show that the effects are equal.

The ownership interaction remains negative and insignificant when using all 384 balanced stores or treating temporary closures as zero. A wage-gap interaction is also insignificant. Full sensitivity estimates are in outputs/tables/robustness.csv. Ownership is not randomly assigned, so the extension is exploratory.

6 Limitations and conclusion

The data have only two survey waves and may contain reporting errors. State-specific shocks could affect the DID, and store-level standard errors do not resolve policy inference with only two states. The sample covers existing restaurants in four chains, so the findings should not be applied to every industry or to new store entry.

The project recovers the main positive employment comparisons in the paper. It does not provide clear evidence that this policy reduced employment in the surveyed restaurants. The ownership extension is inconclusive. These results do not establish that minimum-wage increases always raise employment.

AI use and reproduction

ChatGPT Codex assisted with interpreting the paper, developing and debugging code, preparing figures and drafting the report. Checks included source-table comparisons, sample checks, an independent baseline regression and a clean-input rerun. The full record is in AI_USE_DISCLOSURE.md. Responsibility for reviewing the submitted work remains with the author.

After providing the documented data, install requirements.txt and run python code/run_all.py. This regenerates all tables, figures, processed samples and this report. Row-level data are supplied separately to the teaching team rather than published in the repository.

References

Card, D., and A. B. Krueger (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. American Economic Review, 84(4), 772-793. https://davidcard.berkeley.edu/papers/njmin-aer.pdf

ECO6067 project instructions, Package 01 README and data dictionary, and the teaching-team repository example. Accessed 18 October 2026.